

The brain ascends reward gradients to learn in continuous motor spaces

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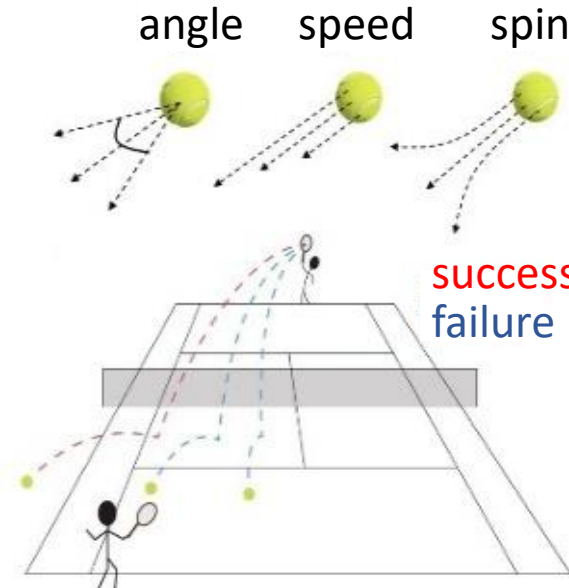
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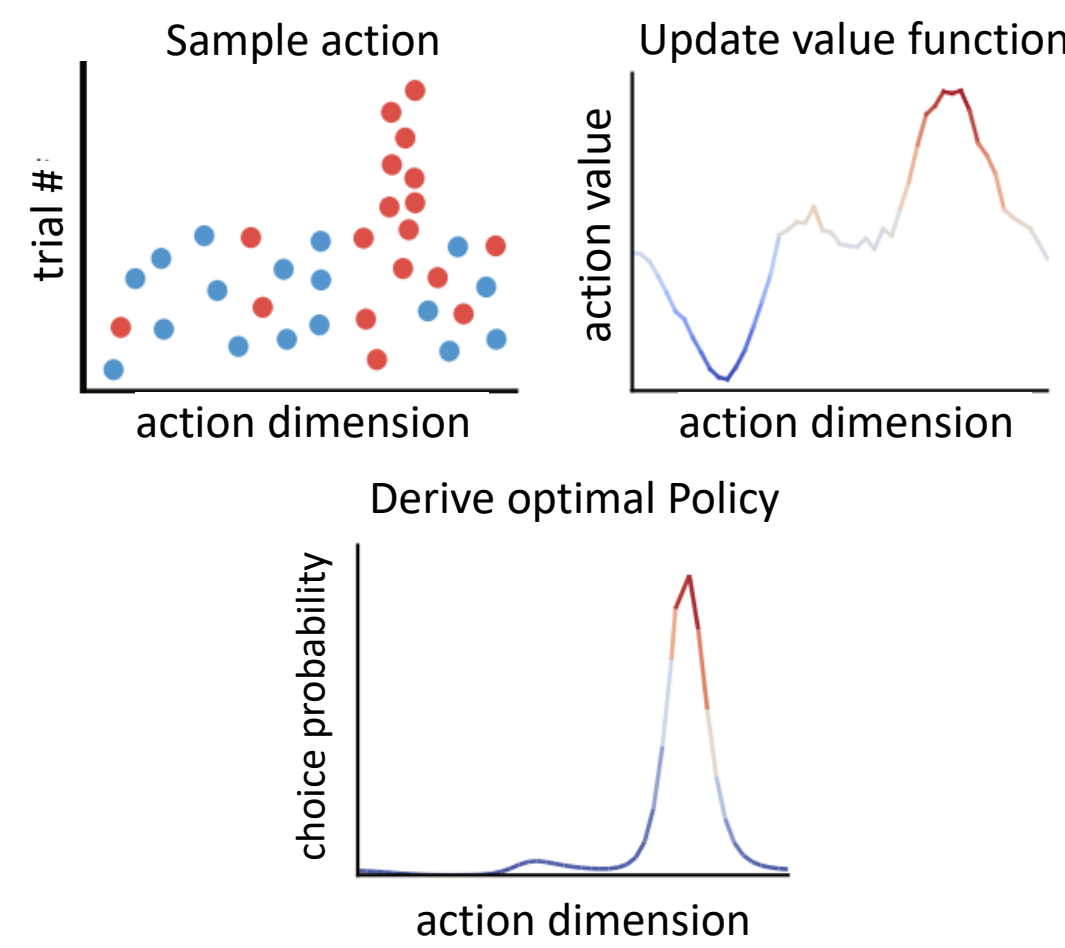
MOTIVATION

Our brain learns complex motor skills like a tennis serve or tying shoelaces through trial-and-error reinforcement learning. Actions are optimized based on reward feedback from the environment.

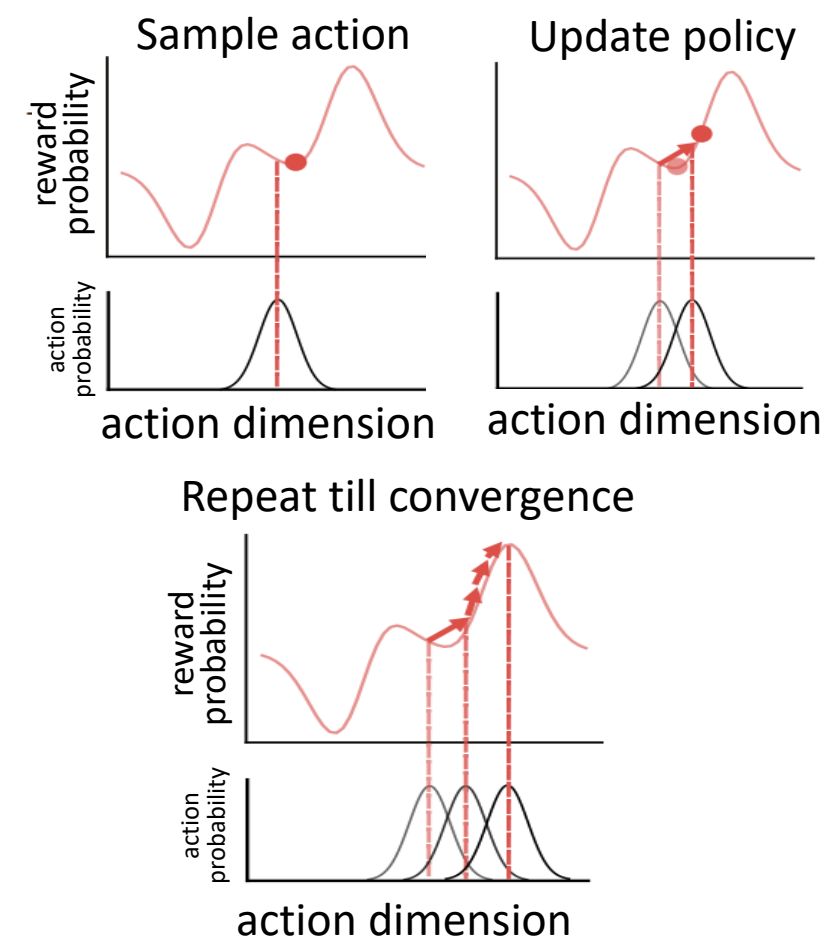
Motor learning requires the brain to explore high dimensional continuous action spaces.



Value-based strategy



Gradient-based strategy



Does the brain follow a reward gradient when learning in continuous motor spaces?

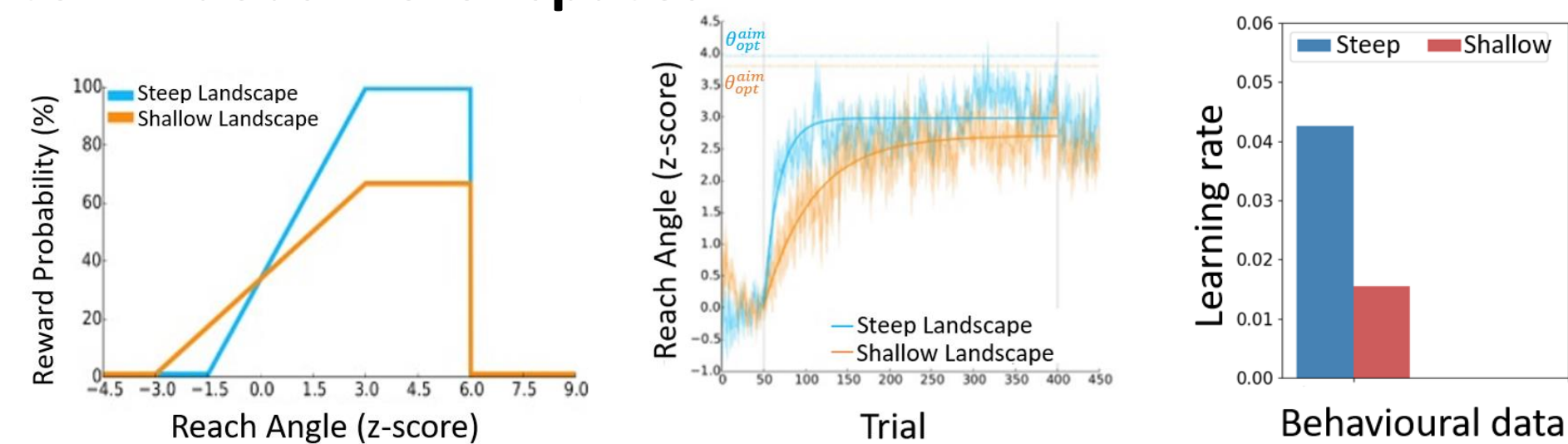


Figure adapted from Cashaback JGA, Lao CK, Palidis DJ, Coltman SK, McGregor HR, Gribble PL. The gradient of the reinforcement landscape influences sensorimotor learning. *PLoS Comput Biol*. 2019

Can modulation of learning rate by reward gradients be used to infer the RL strategy employed by the brain?

METHODS

We simulated each RL strategy on steep and shallow reward gradients.

Reinforcement learning strategies

On trial t sample action a_t to obtain binary reward $r_t = 0/1$

Value-based strategies

Divide the action space into 10 discrete bins. Sample the midpoint m_k of bin k , with gaussian motor noise σ_m

Q-learning

i^{th} bin has associated Q value Q_i

Action policy:

$$\pi_i(t) = \frac{e^{Q_i/\tau}}{\sum_{i=1}^n e^{Q_i/\tau}} \rightarrow a_t \sim N(m_k, \sigma_m^2)$$

Learning update:

$$Q_k(t+1) = Q_k(t) + \alpha(r_t - Q_k(t))$$

Thompson sampling

i^{th} bin has an associated prior and posterior α_i and β_i

Action policy:

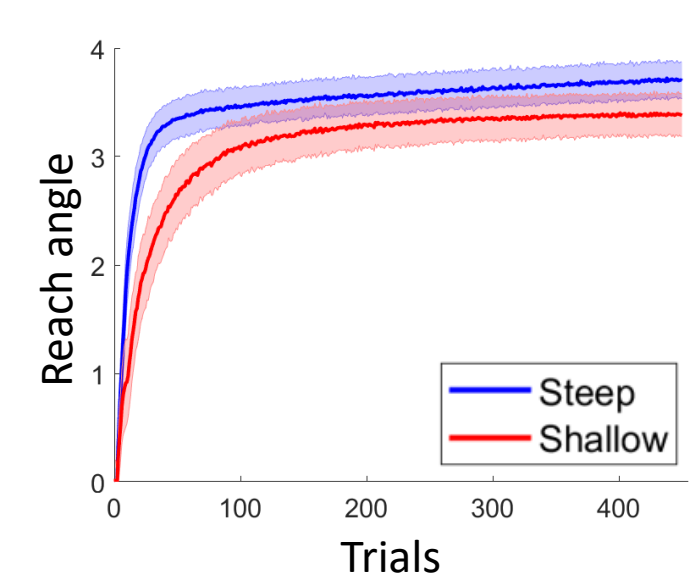
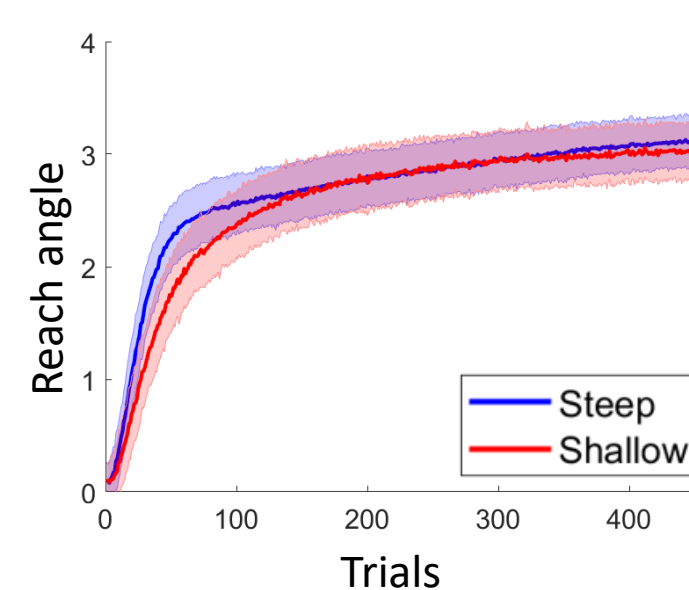
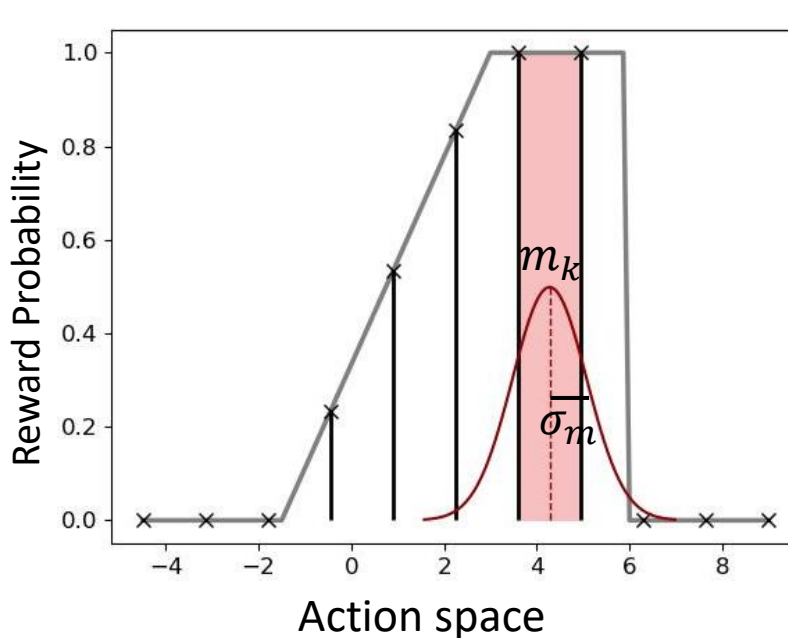
$$\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$$

$$k = \text{argmax}_i (\theta_i) \rightarrow a_t \sim N(m_k, \sigma_m^2)$$

Learning update:

$$\text{If } r_t = 1 \rightarrow \alpha_k(t+1) = \alpha_k(t) + \delta$$

$$\text{If } r_t = 0 \rightarrow \beta_k(t+1) = \beta_k(t) + \delta$$



Policy gradient method:

Sample with mean μ_t and variance comprising of 2 components - σ_e (exploratory variability) and σ_m (motor noise)

Action policy:

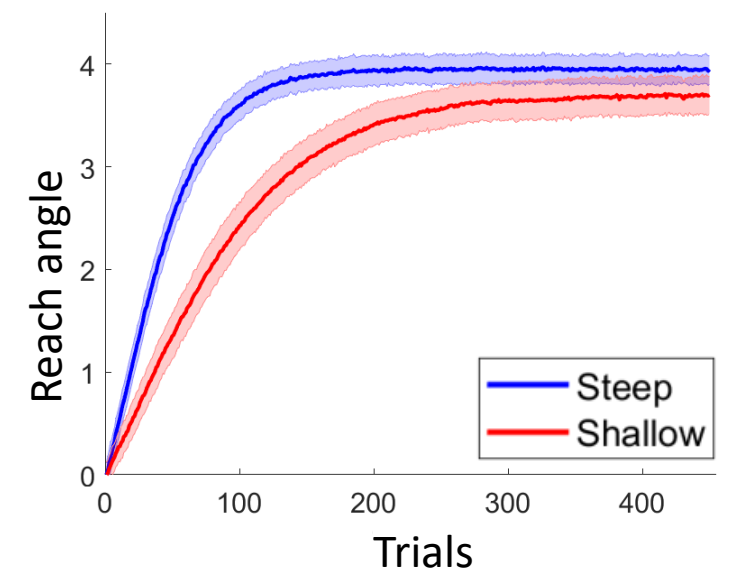
$$a_t \sim N(\mu_t, \sigma_e^2 + \sigma_m^2)$$

Learning update:

$$\delta_t = r_t - \bar{r}_t \text{ and } \bar{r}_{t+1} = \bar{r}_t + \alpha_r \delta_t$$

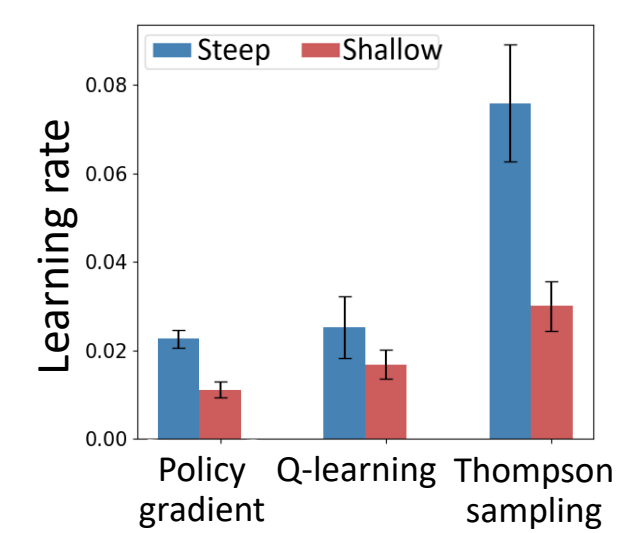
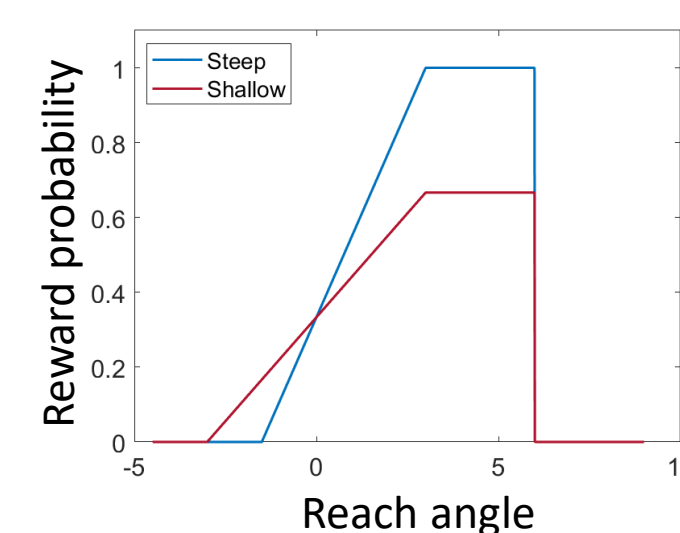
$$\text{Gradient} = \delta_t \varepsilon_t, \text{ where } \varepsilon_t = a_t - \mu_t$$

$$\mu_{t+1} = \mu_t + \alpha_\mu \delta_t \varepsilon_t$$

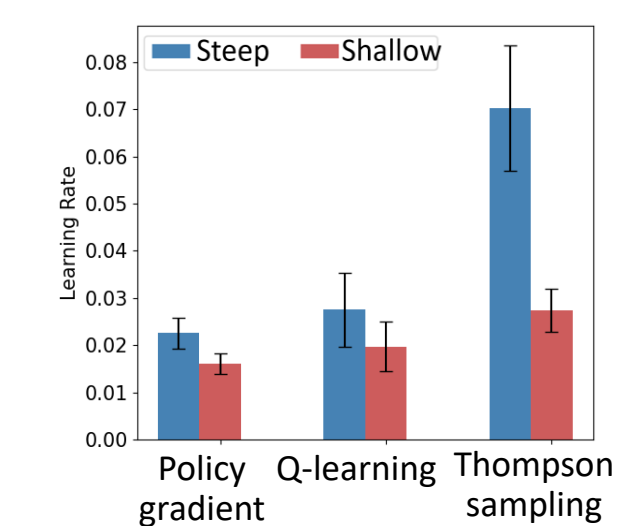
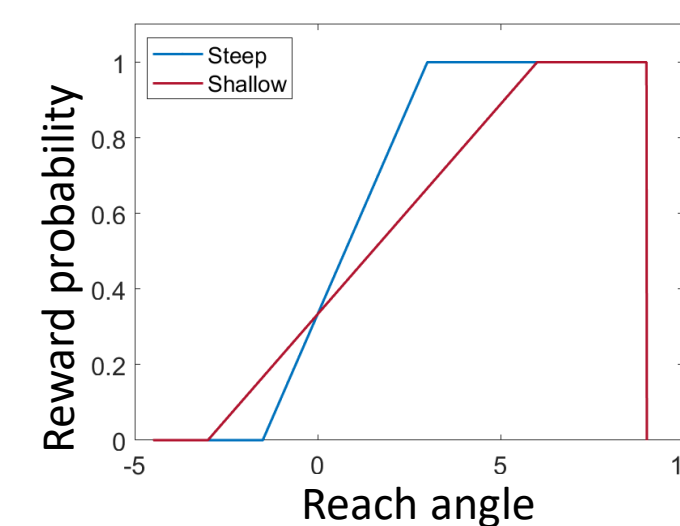


RESULTS

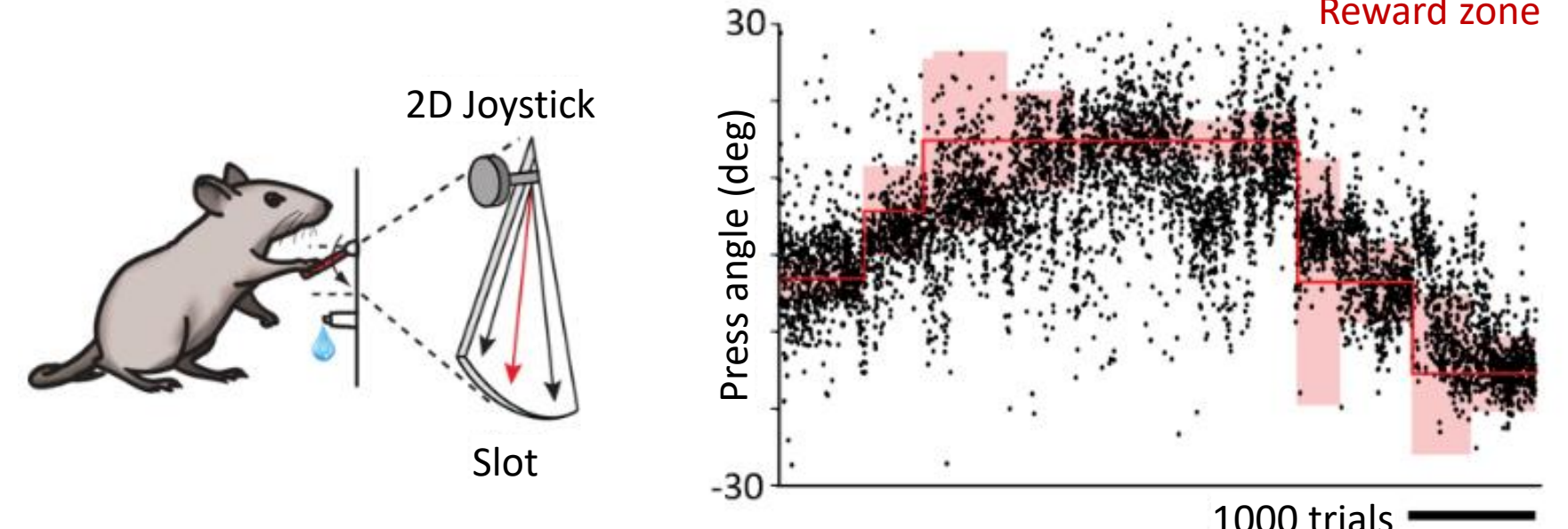
Both value-based and policy gradient methods learn faster on steeper reward gradients



Differences persist in normalized reward landscapes

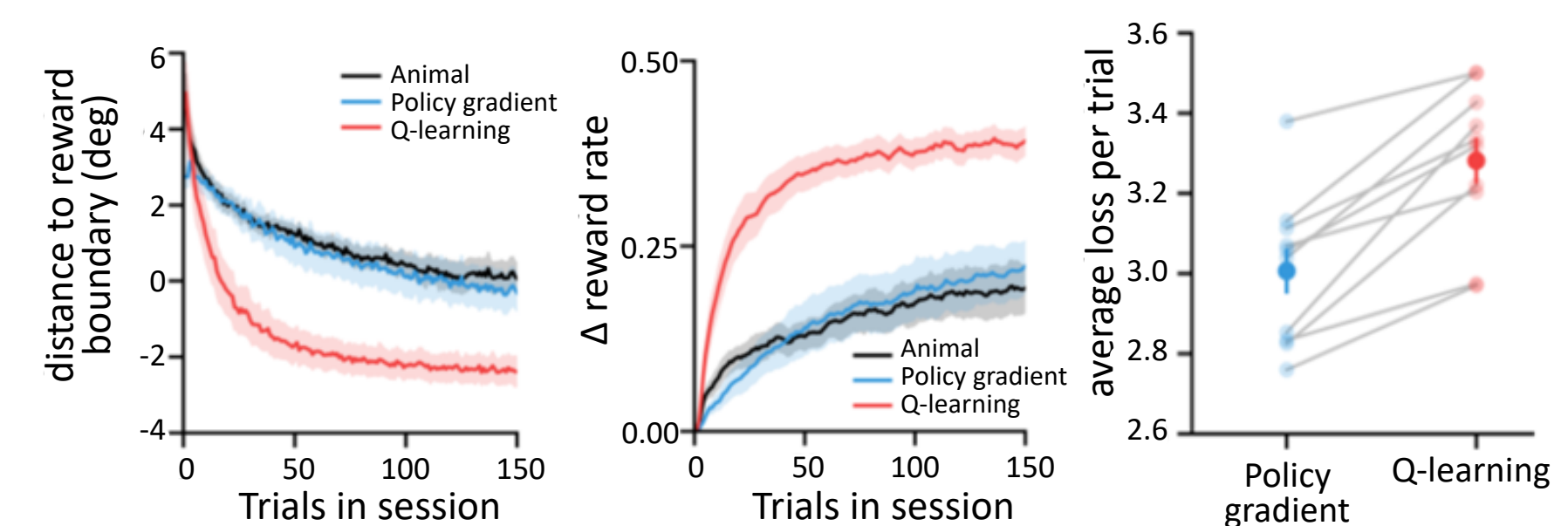


Large behavioral datasets help us infer learning strategies at the scale of individual trials

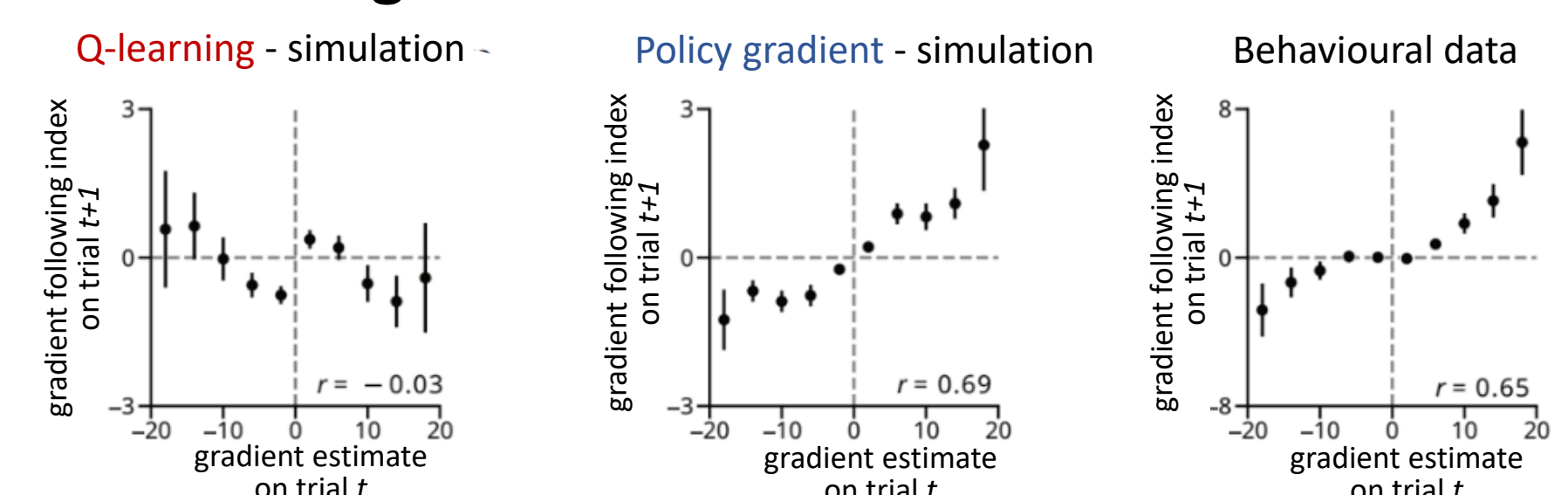


Ashesh K. Dhawale, Yohsuke R. Miyamoto, Maurice A. Smith, Bence P. Ölveczky, Adaptive Regulation of Motor Variability, *Current Biology*, Volume 29, Issue 21, 2019

Policy gradient methods fit behavioral data better



Evidence of gradient ascent at scale of individual trials



CONCLUSION

Our brains learn complex motor skills with great efficiency compared to artificial agents. Our results show that the brain employs gradient ascent algorithms that rely on local exploration for motor learning. Uncovering the brain's strategies can help us understand the principles of efficient learning in continuous action spaces, which can potentially be applied to solve motor learning challenges in the field of robotics.